On the normalization of team performance in mixed rowing relays

Lukas Johannes Splitthoff[1,2], Johan Lysemark[2], Oscar Gyllensten[2]

[1]MC2, Chalmers University of Technology, Göteborg, Sweden

[2]Mölndals Roddklubb, Mölndal, Sweden

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We present a method for normalizing the results of a mixed-gender indoor-rowing relay competition. The Råda Rowing Challenge 2026 involved ten five-person relay teams with different gender compositions, making direct comparison of raw distances problematic. We introduce a power-based normalization scheme in which each team's average power output is compared to a composition-weighted reference derived from world indoor-rowing records. We show that this approach is physically better motivated than a linear time-based normalization, because rowing performance scales as the cube of the rowing velocity. The method is applied to the competition results, and we discuss how the ranking differs from a naive distance ranking.

I. INTRODUCTION

The Råda Rowing Challenge (RRC) is an indoor rowing competition organized by Mölndals Roddklubb, held annually at RådaRum in Mölnlycke, Sweden. The 2026 edition took place on 7 March 2026 and included individual events as well as a 30-minute mixed-relay format [1]. In the relay, each team consists of five rowers who take turns on a Concept2 RowErg ergometer; the combined distance covered by all five during the 30-minute window is the team's result.

Ten teams from clubs across the Gothenburg region entered the relay: Chalmers Roddklubb (two teams), Jönköpings Roddsällskap (two teams), Halmstads Roddklubb, Mölndals Roddklubb (two teams), Göteborgs Roddklubb, Råda Crew, and CTH-GU. Team compositions ranged from all-female (Göteborgs Roddklubb) to four-men-one-woman (Chalmers Roddklubb – Team 2), reflecting the mixed nature of the competition.

Because men and women differ systematically in their power output on the ergometer, a simple comparison of total relay distances would favour teams with more male rowers. To enable a fair ranking across different gender compositions, we develop a normalization procedure described in the following sections. Section II introduces the ergometer power formula. Section III summarizes relevant world records and the performance gap between genders. Section IV presents two normalization approaches and argues for power-based normalization. Section V reports the competition results.

II. THE ERGOMETER POWER FORMULA

II.1. Derivation

The Concept2 ergometer flywheel decelerates between drive phases due to air resistance. The drag force scales quadratically with the flywheel angular velocity, which is proportional to the handle velocity v of the rower. Since power equals force times velocity, the average power delivered to the flywheel is

$$P = c\,v^3, \tag{1}$$

where $v = d/T$ is the average rowing velocity (distance d in metres, time T in seconds) and c is the flywheel drag constant. For the Concept2 RowErg [2],

$$c = 2.8\,\mathrm{W\,s^3/m^3}. \tag{2}$$

Equation (1) can equivalently be expressed using the *500-metre split time* t_{500}, the time to row 500 m at the current average pace:

$$P = 2.8\left(\frac{500}{t_{500}}\right)^3, \qquad t_{500} = \frac{500\,T}{d}. \tag{3}$$

A faster split (smaller t_{500}) gives higher power, as physically expected. For example, the men's reference pace (5000 m in 939 s) corresponds to $t_{500} = 500 \times 939/5000 \approx 94$ s (split 1:34 min).

II.2. Power curve

Figure 1 shows P as a function of t_{500} over the range typical for competitive indoor rowing. The strong nonlinearity – power grows as the *cube* of velocity – is the central motivation for using power rather than velocity or split time as the performance metric.

III. REFERENCE PERFORMANCE

III.1. World indoor rowing records

Table 1 lists world indoor rowing records for 2000 m and 5000 m on the static Concept2 RowErg, as reported at the 2026 World Rowing Virtual Indoor Championships [3].

III.2. Performance spread and the gender gap

At the competitive level, 2000 m times for male club and national-team rowers typically span 5:30–6:10, corresponding

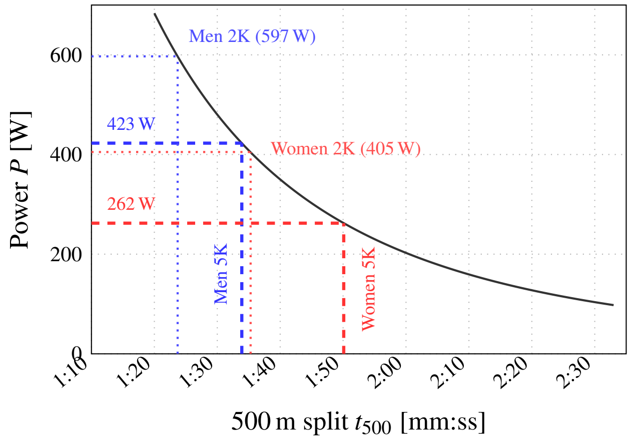

Figure 1: Ergometer power vs. 500 m split time, Eq. (3). Dashed lines: reference performance levels used in this study (5000 m world records from the 2026 WRVIC). Dotted lines: 2000 m world records (men: O. Zeidler 2026; women: B. Mooney 2021).

Table 1: World indoor rowing records (Concept2 RowErg, open category). Power computed via Eq. (3).

Dist.	Gender	Time	P [W]	Athlete (year)
2000 m	Men	5:34.7	597	O. Zeidler (2026)
2000 m	Women	6:21.1	405	B. Mooney (2021)
5000 m	Men	15:39	423	WRVIC (2026)
5000 m	Women	16:49.4	341	G. Rowe (2026)

to 470 W–635 W. Female athletes at comparable relative levels achieve 6:10–7:00 (270 W–395 W). The spread is broadly similar in relative terms for both groups, yet the absolute power gap between the two reference levels in Table 1 is about 61 % (423 W vs. 262 W for 5000 m). In split time, the same gap is only 17 % (1:33.9 vs. 1:50.1 per 500 m). This asymmetry is a direct consequence of the cubic relation $P \propto v^3$: a moderate advantage in speed translates into a much larger advantage in power. As a result, the gender composition of a relay team has a strong systematic effect on the expected power output, motivating the normalization approach described in the next section.

IV. NORMALIZATION

Let team k comprise n_m male and n_f female rowers, with $n = n_m + n_f$. During the 30-minute relay ($T = 1800\,\mathrm{s}$) the team covers a combined distance D_k, giving an average team power

$$P_k^{\mathrm{act}} = 2.8 \left(\frac{D_k}{T}\right)^3. \tag{4}$$

IV.1. Time-based normalization

A natural first approach is to compare D_k to the distance a reference team of the same composition would cover at the gender-specific reference pace. Defining $v_m = d_{\mathrm{ref}}/T_m^{\mathrm{ref}}$ and $v_f = d_{\mathrm{ref}}/T_f^{\mathrm{ref}}$ as the male and female reference velocities, the expected relay distance is

$$D_k^{\mathrm{ref,t}} = \frac{T}{n}(n_m v_m + n_f v_f), \tag{5}$$

and the time-normalized score $S_k^{(t)} = D_k / D_k^{\mathrm{ref,t}}$ is dimensionless and equals 1.0 for a team rowing exactly at the composition-weighted reference pace.

IV.2. Power-based normalization

We propose instead to compare power outputs directly. The reference power for team k is the composition-weighted mean of individual reference powers:

$$P_k^{\mathrm{ref}} = \frac{n_m P_m + n_f P_f}{n}, \tag{6}$$

where P_m and P_f are the male and female reference powers (from Table 1, rows 3–4). The power-normalized score is

$$S_k = \frac{P_k^{\mathrm{act}}}{P_k^{\mathrm{ref}}}. \tag{7}$$

IV.3. Why power normalization is preferred

The two methods treat the nonlinearity of $P \propto v^3$ differently. Time normalization is linear in velocity: it credits a rower who rows 10 % above reference with a 10 % bonus, regardless of the absolute speed difference. Power normalization is cubic: the same 10 % velocity bonus corresponds to $(1.10)^3 - 1 \approx 33\,\%$ more power, and this full advantage is reflected in the score.

More importantly, the Concept2 ergometer *measures* power – it is the physically native quantity of the instrument. Averaging power across team members, Eq. (6), correctly weights each rower's physical contribution. By contrast, averaging reference velocities and then converting to a reference distance, Eq. (5), applies a linear average where the true relationship is cubic, underestimating the advantage of having high-power rowers on the team.

Power-based normalization also has a natural interpretation: a score of $S_k = 1.0$ means the team delivered exactly the power expected for their composition; $S_k > 1$ means they exceeded the reference, and $S_k < 1$ that they fell short.

V. RESULTS

V.1. Reference parameters

We use the 5000 m world records from the 2026 WRVIC (Table 1, rows 3–4) as reference performance:

$$T_m^{\mathrm{ref}} = 939\,\mathrm{s}\ (15{:}39), \quad P_m = 423\,\mathrm{W}, \tag{8}$$
$$T_f^{\mathrm{ref}} = 1101\,\mathrm{s}\ (18{:}21), \quad P_f = 262\,\mathrm{W}. \tag{9}$$

The 5000 m distance is approximately the total relay contribution of one team member over the 30-minute race at a competitive club pace.

V.2. *Race results*

Table 2 presents the ten teams, their compositions, the total relay distances, reference and actual power, and normalized scores computed via Eqs. (4)–(7). Figure 2 visualizes the same data.

V.3. *Discussion*

The power-normalized ranking differs substantially from a raw-distance ranking for several teams. CTH-GU covered the greatest total distance (9.511 km) yet ranks third, held back by a relatively strong expected composition (3M/2F, $P_{\mathrm{ref}} = 359$ W). Conversely, Göteborgs Roddklubb – the only all-female team – covered only 8.575 km but ranks second overall, having substantially exceeded their (lowest possible) reference power of 262 W.

Chalmers Roddklubb – Team 2 illustrates the penalty for a heavy-male composition: despite covering 9.095 km (fourth largest raw distance), they rank seventh after normalization, because their four-man composition sets a demanding reference bar of 391 W.

The cubic power law amplifies small pace differences into large score differences. The rank-1 team (score 1.171) averaged approximately 1:39.5/500 m against a reference of 1:44.2/500 m – a pace advantage of only $\sim$5%, yet a power advantage of $\sim$17%.

VI. CONCLUSION

We have presented a power-based normalization method for mixed-gender indoor rowing relay competitions. The fundamental argument is that rowing power scales as the cube of velocity, Eq. (1), so small velocity differences between genders translate into large power differences. Normalizing by power rather than by distance or time therefore provides a physically motivated and instrument-native measure of relative team performance.

Applied to the RRC 2026, the normalized ranking differs substantially from the raw-distance ranking for several teams. The correction is particularly significant for all-female and predominantly male teams, illustrating the importance of accounting for gender composition in mixed relay events.

The normalization algorithm is released as the open-source Python package `ergrace` [4] and can be applied to future editions of the competition or similar mixed-relay formats.

DATA AVAILABILITY

The race data, normalization code, and the source of this paper are openly available at `https://github.com/lukassplitthoff/ergrace-normalizer`.

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AUTHOR CONTRIBUTIONS

LJS wrote the paper and developed the power normalization. JL assisted with data acquisition and analysis. OG acquired the data and supervised the project.

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Table 2: RRC 2026 relay results ranked by normalized score. n_m/n_f: male/female rowers. P_{ref}: composition-weighted reference power, Eq. (6). P_{act}: team average power, Eq. (4). Score $= P_{\mathrm{act}}/P_{\mathrm{ref}}$.

Rank	Team	Lane	Comp.	Dist. [km]	P_{ref} [W]	P_{act} [W]	Score
1	Chalmers Roddklubb – Team 1	8	2M/3F	9.268	326	382	1.171
2	Göteborgs Roddklubb	7	0M/5F	8.575	262	303	1.155
3	CTH-GU	10	3M/2F	9.511	359	413	1.152
4	Jönköpings Rodds. – Team 2	5	2M/3F	9.021	326	353	1.080
5	Mölndals Roddklubb – Team 1	4	2M/3F	8.744	326	321	0.983
6	Mölndals Roddklubb – Team 2	6	1M/4F	8.439	294	289	0.981
7	Chalmers Roddklubb – Team 2	1	4M/1F	9.095	391	361	0.924
8	Jönköpings Rodds. – Team 1	2	3M/2F	8.660	359	312	0.870
9	Råda Crew	9	3M/2F	7.733	359	222	0.620
10	Halmstads Roddklubb	3	2M/3F	6.956	326	162	0.495

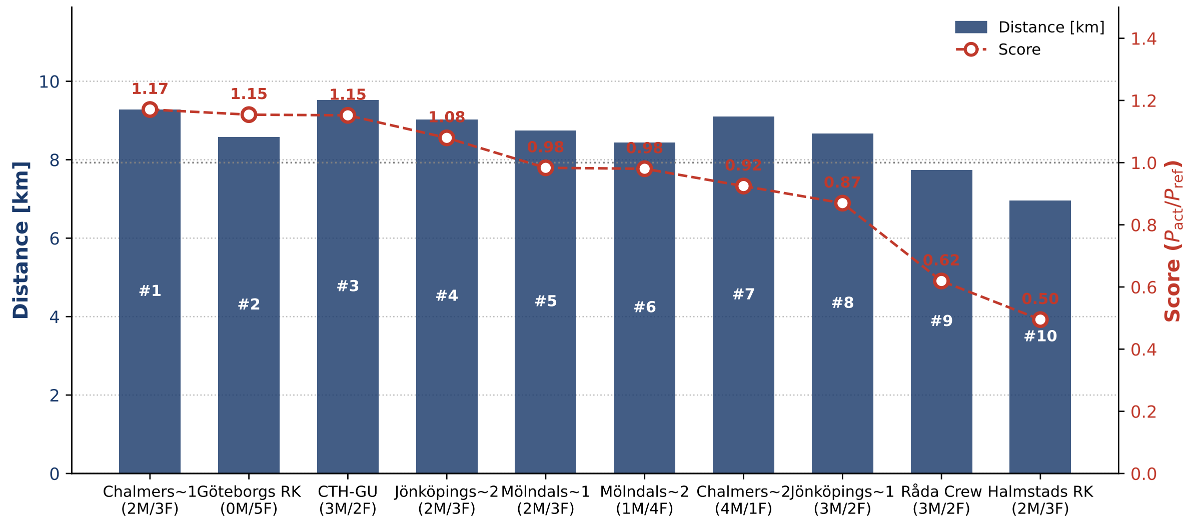

Figure 2: RRC 2026 relay results. Bars (left axis): total relay distance per team. Markers (right axis): normalized score. Teams ordered by score (best to worst, left to right). The dashed line marks score $= 1.0$ (reference performance level). Generate this figure by running `generate_figures.py`.